Final Assignment XAI

Jorrit Roobol

March 2026

1 Motivation and Context

The use of machine learning in the financial sector has grown rapidly over the last few years. Banks now use complex models to decide whether a person is eligible for a credit card or a loan. While these models are very accurate, they are often seen as black boxes. This means that when a person is rejected, the bank cannot easily explain why that decision was made. This lack of transparency is a major problem for both the customer and the bank.

From a legal perspective, explaining these decisions is becoming a requirement. The General Data Protection Regulation in Europe mentions that individuals have a right to receive meaningful information about the logic that is used in automated decisions. More recently, the EU AI Act has classified credit scoring as a high-risk use of AI. This means that systems used for these purposes must be transparent and explainable. My project aims to look at this need by comparing different ways to provide these explanations using the UCI Credit Card dataset [1].

The goal of this project is to see how different XAI methods can help a applicant who did not get their loan. I am comparing a traditional method that shows feature importance with a newer method that provides counterfactual examples. By doing this, I want to find out which approach is more helpful for a regular person who wants to know what they can do to improve their financial situation and get their loan in the future.

2 Methodology

In this project, I used two different explainable AI methods to analyze a Random Forest model trained on the credit card default dataset [1]. The first method is SHAP, which represents a global and local feature importance approach learned in the course. The second method is DiCE, which provides counterfactual explanations. These two methods were chosen because they offer different types of information.

SHAP is based on the concept of Shapley values from cooperative game theory [2]. It works by looking at every possible combination of features to determine how much each individual feature contributes to the final prediction.

In the context of a loan rejection, SHAP can show that a person's payment history had a much larger impact on the decision than their age or education level. This gives a fair and mathematically grounded way to assign credit or blame to specific parts of a person's financial profile. Because calculating every combination asks way to much power from my computer, I used a sample so I could run the tests more quickly, and hopefully still be accurate.

The second method, DiCE, stands for Diverse Counterfactual Explanations [3]. Unlike SHAP, which explains the current state of the model, DiCE looks at what would happen if the input data were different. It generates a set of counterfactual examples, which are versions of the applicant that are very similar to the original but have a different outcome, such as being approved instead of rejected. This method is useful for credit scores because it shows you wat you can do to get a better credit score and get approved. Instead of knowing that their income is a negative factor, you get a tip, such as increasing their savings by a certain amount to change the model's decision.

To ensure the explanations are useful, I focus on three evaluation metrics. First, I check for validity, ensuring that the changes suggested by DiCE actually result in a different model prediction. Second, I look at sparsity, which measures how many features are changed in an explanation. I would say a lower number of changes is better because you have to focus on less. Finally, I consider proximity, which means that I want the changes that have to be made to be as small as possible, so it will not be an impossible task.

3 Results and Analysis

The results from my experiment show a clear difference between how SHAP [2] and DiCE [3] explain the same model decision. The Random Forest model achieved an accuracy of 0.82 on the test set, which suggests the explanations are based on a relatively reliable model.

In the SHAP analysis, the payment status features (such as PAY_0) showed the highest feature importance across the sampled test cases. This provides a global understanding of the model, indicating that the bank's AI prioritizes recent payment behavior when deciding on credit defaults. While this is useful for the bank to verify the model's logic, it does not tell a rejected applicant what specific steps they should take.

The DiCE counterfactuals provided more specific information. For one rejected applicant, the system generated three different scenarios that would change the decision to an approval. However, an analysis of these results revealed a significant pitfall regarding actionability. One counterfactual suggested that the applicant would be approved if their age was lower. Since age is something that you can not change, this advice is not helpful in a real-world scenario. Another counterfactual suggested a very high payment amount that might not be financially feasible for the user. These findings highlight that while counterfactuals are more direct than feature importance scores, they must be constrained to only include realistic and ethical changes.

4 Pitfalls and Biases

While explainable AI aims to make machine learning models more transparent, there are several significant pitfalls and biases that must be considered. One major pitfall encountered in this project relates to the feasibility of counterfactual explanations. As seen in the results, the DiCE method sometimes suggested changes to features that you can not change, such as the age of the applicant. In a real financial application, providing an explanation that suggests a user should be younger to receive a loan is not only useless but also highlights potential age discrimination within the model. This is a common issue in counterfactual generation [3] where the algorithm prioritizes finding the mathematically closest point to the decision boundary without considering whether that change is realistic or even possible.

Another technical pitfall is the issue of feature correlation. Both SHAP and DiCE often treat features as independent variables. However, in the credit card dataset [1], features like bill amount and previous payment amounts are highly correlated. If an explanation suggests decreasing a bill amount without acknowledging the impact on the user's overall balance or payment history, the advice may be financially impossible. This lack of causal awareness can lead to a loss of trust from the applicant, who might feel the AI does not understand the reality of their financial situation.

Regarding bias, the dataset itself may contain historical biases that the Random Forest model has learned. For example, if past lending decisions were biased against certain genders or education levels, the model will likely replicate those patterns. While SHAP [2] can help identify if the model is heavily weighting a protected characteristic like sex or marriage status, it does not solve the bias that was formed. Furthermore, there is a risk of "explanation bias," where an applicant might trust a model more, only because the explanation is clear and easy to understand, even though it might not be the right explanation or if the model is unfair or inaccurate.

Finally, the computational cost of these methods represents a practical pitfall. As seen during the implementation, calculating SHAP values for the entire dataset was not possible on my computer or other standard computers. This suggests that while these methods work well in a research setting, they may face issues when implemented in a way larger banking system that needs to provide millions of explanations in real-time.

5 Conclusion

This project shows that there is no "perfect" version of explainable AI that works for every situation, especially in sensitive areas like bank lending. By comparing SHAP [2] and DiCE [3] using the credit card dataset, I found that different methods help different people. SHAP is very useful for bank experts and data scientists who need to check if the model is working correctly on a large scale. It gives a solid, mathematical look at which factors matter most.

However, for a regular customer, these scores can feel a bit too technical and abstract.

In contrast, DiCE is much better for the individual applicant because it provides "counterfactuals." These are simple "what-if" examples that show exactly what a person needs to change to get approved. While this is more intuitive for a normal person, these models have a major weakness: they often suggest impossible or unfair changes, like telling someone to change their age or gender to get a loan.

The results of this study suggest that the best way forward for banks is a hybrid approach. A successful system should combine the reliability of SHAP with the helpful advice of DiCE. Crucially, developers must add "feasibility constraints" to these models. These are rules that stop the AI from suggesting changes to things a person cannot control. By doing this, banks can ensure that the advice they give to customers is not only mathematically correct but also legal and actually possible to achieve.

Code Availability The Python source code, dataset, and implementation details for this project are publicly available on GitHub at: `https://github.com/JoJoJorrit/XAI_Final_assignment`

References

[1] Yeh, I. C., & Lien, C. H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473-2480.

[2] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

[3] Mothilal, R. K., Sharma, A., & Tan, C. (2020). Explaining machine learning classifiers through diverse counterfactual explanations. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 607-617.